• Limit vs subsequences

Theorem

Let $\{a_n\}$ be a sequence. Then

$$\lim_{n\to\infty} a_n = L \quad \textit{if and only if} \quad \lim_{n\to\infty} a_{2n-1} = \lim_{n\to\infty} a_{2n} = L.$$

Example

Show that $\{a_n\}$ converge, where $a_n = \begin{cases} \frac{2k}{2k+1} & \text{if } n = 2k, \\ e^{1/(2k-1)} & \text{if } n = 2k-1. \end{cases}$

Solution. We have

$$\lim_{n\to\infty} a_{2n} = \lim_{n\to\infty} \frac{2n}{2n+1} = 1 \text{ and } \lim_{n\to\infty} a_{2n-1} = \lim_{n\to\infty} e^{1/(2n-1)} = 1.$$

So the given sequence $\{a_n\}$ converges and $\lim_{n\to\infty} a_n = 1$.

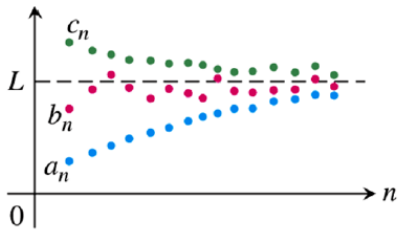
• Squeeze Theorem

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be three sequences, satisfying conditions

- There is $n_0 \in \mathbb{N}$, such that for every $n \geq n_0$, $a_n \leq b_n \leq c_n$,
- $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$.

Then the sequence $\{b_n\}$ also converges and $\lim_{n\to\infty} b_n = L$.

Graphically, $a_n \leq b_n \leq c_n$, for $n \geq n_0$ & $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$



To apply Squeeze Theorem to the sequence $\{b_n\}$, we have to find two sequences $\{a_n\}$ and $\{c_n\}$ so that $\begin{cases} a_n \leq b_n \leq c_n, & \text{for all } n \geq n_0 \\ \lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} c_n = L. \end{cases}$

Example

Does $\lim\limits_{n\to\infty} \dfrac{\cos n}{n}$ exist? If yes, what is its limit?

Solution. For $n \geq 1$, we have $-1 \leq \cos n \leq 1$, which implies that

$$\frac{-1}{n} \leq \frac{\cos n}{n} \leq \frac{1}{n}.$$

Since $\lim\limits_{n\to\infty} \dfrac{1}{n} = \lim\limits_{n\to\infty} \dfrac{-1}{n} = 0$, by Squeeze Theorem $\lim\limits_{n\to\infty} \dfrac{\cos n}{n} = 0$.

• Bounded Sequences

Definition

A sequence $\{a_n\}$ is said to be bounded if there are numbers M and m such that

$$m \leq a_n \leq M, \quad \text{for every } n.$$

Numbers m and M are called a lower bound and an upper bound of the sequence $\{a_n\}$.

For example, the sequence $\{(-1)^n\}_{n=1}^{\infty}$ is bounded ($m = -1$, $M = 1$).

• Boundedness of a convergent sequence

The above example shows that a bounded sequence may not be convergent. However, a convergence property is stronger.

Theorem ("convergence" $\Longrightarrow$ "boundedness")

If a sequence $\{a_n\}$ converges, then it is bounded.

- **Monotone sequences**

Definition

(a) A sequence $\{a_n\}$ is said to be increasing for $n \geq n_0$ if $a_n \leq a_{n+1}$ for every $n \geq n_0$ (where n_0 is some integer).

(b) A sequence $\{a_n\}$ is said to be decreasing for $n \geq n_0$ if $a_n \geq a_{n+1}$ for every $n \geq n_0$ (where n_0 is some integer).

(c) A sequence $\{a_n\}$ is said to be monotonic if it is either increasing or decreasing.

- **About monotone and bounded sequences**

Monotone Convergence Theorem

Every bounded, monotonic sequence is convergent.

Remarks.

- This theorem is used to show the existence of limit, but it does not give the value of the limit.
- The verification of the sequence being bounded and monotone usually involves *mathematical induction*.

• Fitting with a Function

There is a good way to work on sequences using fitting functions.

Definition

Let $\{a_n\}$ be a sequence. A function $f : \mathbb{R} \to \mathbb{R}$ satisfying $a_n = f(n)$, for all $n \geq n_0$ (n_0 is some integer), is called a fitting function for $\{a_n\}$.

• Theorem about a fitting function

The following assertions are true.

(a) If $\lim_{x\to\infty} f(x) = L$, then $\lim_{n\to\infty} a_n = L$.

(b) If $\lim_{x\to\infty} f(x) = \pm\infty$, then $\lim_{n\to\infty} a_n = \pm\infty$.

Note. When a sequence $\{a_n\}$ can be fitted by a function $f(x)$, we can apply L'Hospital's Rule to $\lim_{x\to\infty} f(x)$.

Outline

1. Sequences, Convergence, Limits
2. Infinite Series
3. Properties of Series, Tests for Convergence
4. Power Series, Taylor & Maclaurin Series

Definition

Let $\{a_n\}$ be a sequence. An infinite series is a formal expression (of an "infinite sum") of the form

$$a_1 + a_2 + a_3 + \cdots, \quad \text{denoted as } \sum_{n=1}^{\infty} a_n.$$

Sometimes, we also represent an infinite series as $\sum_{n=0}^{\infty} a_n$.

For example, for the series $\sum_{n=1}^{\infty} \frac{1}{(n-1)!}$, we prefer to write it as $\sum_{n=0}^{\infty} \frac{1}{n!}$.

Definition

- For a series $\sum_{n=1}^{\infty} a_n$, the sequence of its partial sums $\{S_k\}$ is defined by

$$S_k = a_1 + a_2 + a_3 + \cdots + a_k \quad (k = 1, 2, \ldots)$$

- The series $\sum_{n=1}^{\infty} a_n$ is said to converge, if the sequence $\{S_k\}$ converges. Moreover, if $\lim_{k\to\infty} S_k = S$, then we say the given series converges to S, and write $\sum_{n=1}^{\infty} a_n = S$. The number S is called the sum of the series.
- If a series does not converge, then it is said to diverge.

Example

Determine whether the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges.

Solution. We first perform a sequence of partial sums $\{S_k\}$:

$$S_k = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{k(k+1)} \quad (k \geq 1).$$

Note that $\frac{1}{m(m+1)} = \frac{1}{m} - \frac{1}{m+1}$, for each $m \geq 1$. Then we have

$$S_k = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{k} - \frac{1}{k+1}\right) = 1 - \frac{1}{k+1}.$$

Since $\lim_{k \to \infty} S_k = \lim_{k \to \infty} \left(1 - \frac{1}{k+1}\right) = 1$, we conclude that the given series converges and its sum is 1.

Geometric series

A geometric series is a series

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + ar^3 + \cdots + ar^n + \cdots,$$

where $a \neq 0$ is a constant number, and for any $n \geq 0$, $\dfrac{a_{n+1}}{a_n} = r$ is also a constant number (known as the common ratio r).

The geometric series $\displaystyle\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + \cdots + ar^n + \cdots$ $(a \neq 0)$ is

(a) convergent if $|r| < 1$ and its sum is $\dfrac{a}{1-r}$;

(b) divergent if $|r| \geq 1$.

We can remember the formula $\boxed{\text{Sum} = \dfrac{\text{First term}}{1 - (\text{Common ratio})}}$.

- **Harmonic series**

The harmonic series is the following series:

$$1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots+\frac{1}{n}+\cdots=\sum_{n=1}^{\infty}\frac{1}{n}.$$

It can be shown that for its sequence of partial sums $\{S_k\}$

$$\lim_{k\to\infty} S_k=\infty.$$

Thus the harmonic series $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n}=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots+\frac{1}{n}+\cdots$ diverges.

Outline

1. Sequences, Convergence, Limits
2. Infinite Series
3. Properties of Series, Tests for Convergence
4. Power Series, Taylor & Maclaurin Series

• Properties of Series

• **Necessary condition of convergence**: If a series $\sum_{n=0}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

The converse is not true (for instance, the harmonic series).

• **Operations of convergent series**: If series $\sum_{n=0}^{\infty} a_n$, $\sum_{n=0}^{\infty} b_n$ converge, then so do the series $\sum_{n=0}^{\infty} Ca_n$ (C is any constant) and $\sum_{n=0}^{\infty}(a_n \pm b_n)$.

Moreover,

$$\sum_{n=0}^{\infty} Ca_n = C\sum_{n=0}^{\infty} a_n \quad \text{and} \quad \sum_{n=0}^{\infty}(a_n \pm b_n) = \sum_{n=0}^{\infty} a_n \pm \sum_{n=0}^{\infty} b_n.$$

The necessary condition for convergence is useful in its contrapositive form: if $\lim_{n\to\infty} a_n$ does not hold, then the series $\sum_{n=0}^{\infty} a_n$ diverges.

• Test for divergence

If $\lim_{n\to\infty} a_n$ does not exist or $\lim_{n\to\infty} a_n \neq 0$, then the series $\sum_{n=0}^{\infty} a_n$ diverges.

Example

(a) $\sum_{n=1}^{\infty}(-1)^n$ diverges, since $\lim_{n\to\infty}(-1)^n$ does not exist.

(b) $\sum_{n=1}^{\infty} n$ diverges, since $\lim_{n\to\infty} n = \infty$ (does not exist).

(c) $\sum_{n=1}^{\infty} \frac{2n+1}{n+2}$ diverges, since $\lim_{n\to\infty} \frac{2n+1}{n+2} = 2 \neq 0$.

- **Alternating series**

An alternating series is a series whose terms are alternately positive and negative.

Example

- $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots$
- $-2 + 4 - 8 + 16 - 32 + \cdots$
- In general, $\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + \cdots \quad (b_n > 0)$

• Test for alternating series

If the alternating series

$$\sum_{n=1}^{\infty}(-1)^{n-1}b_n = b_1 - b_2 + b_3 - b_4 + \cdots \quad (b_n > 0)$$

satisfies $\begin{cases} \lim\limits_{n\to\infty} b_n = 0 \\ b_{n+1} \leq b_n, \text{ for all } n, \end{cases}$ then the series converges.

Example

The alternating harmonic series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ is convergent.

It is clear by the Test for alternating series (in this case, $b_n = \frac{1}{n} \downarrow 0$).

- **A study of non-negative series**

These are series with non-negative terms: $\sum_{n=1}^{\infty} a_n$, $a_n \geq 0$ for all n.

There are special techniques for testing the convergence of such series.

Since $a_n \geq 0$, the sequence of partial sums $\{S_n\}$ is increasing: $S_{n+1} = S_n + a_{n+1} \geq S_n$, for all n, and hence

$$S_1 \leq S_2 \leq S_3 \leq \cdots \leq S_n \leq S_{n+1} \leq \cdots .$$

By the Monotone Convergence Theorem, we have the following result.

A series $\sum_{n=1}^{\infty} a_n$ of non-negative terms converges if and only if its sequence of partial sums $\{S_n\}$ is bounded (from above).

- **Various tests for convergence of non-negative series**

(1) Integral Test: Suppose f is a continuous, positive, decreasing function of x for all $x \geq N$ (N is some integer), and $a_n = f(n)$. Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x)\,dx$ both converge or both diverge.

(2) Comparison Test: Let $\sum a_n$ and $\sum b_n$ be series with non-negative terms. Suppose there is a number N such that $a_n \leq b_n$, for all $n > N$.

- If $\sum b_n$ converges, then $\sum a_n$ also converges.
- If $\sum a_n$ diverges, then $\sum b_n$ also diverges.

(3) Limit Comparison Test: Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N is some integer). Consider a limit $\lim_{n\to\infty} \frac{a_n}{b_n} = c$.

$$\begin{cases} \bullet\ c > 0 \implies \sum a_n \ \& \ \sum b_n \text{ both converge or both diverge} \\ \bullet\ c = 0 \ \& \ \sum b_n \text{ converges} \implies \sum a_n \text{ converges} \\ \bullet\ c = \infty \ \& \ \sum b_n \text{ diverges} \implies \sum a_n \text{ diverges.} \end{cases}$$

(4) p-series: The p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \cdots, + \frac{1}{n^p} + \cdots \ (p \text{ is a real constant})$$

converges if $p > 1$, and diverges if $p \leq 1$.

Exercise

Determine the convergence of the following series.

1. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{1+n^2}.$
2. $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n}{n}.$
3. $\displaystyle\sum_{n=1}^{\infty} \frac{5}{2n^2+4n+3}.$

[Answers]. (1) Convergent; (2) Divergent; (3) Convergent.

• The Ratio & Root Tests

Suppose $\sum a_n$ is a series with positive terms for $n > N$ (N is some integer). Suppose $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \alpha$, $\lim_{n\to\infty} \sqrt[n]{a_n} = \beta$. Then

1. $\begin{cases} \bullet\ \alpha < 1 \implies \sum a_n \text{ converges} \\ \bullet\ \alpha > 1 \text{ or } \alpha = \infty \implies \sum a_n \text{ diverges} \\ \bullet\ \alpha = 1 \implies \text{no conclusion.} \end{cases}$
2. $\begin{cases} \bullet\ \beta < 1 \implies \sum a_n \text{ converges} \\ \bullet\ \beta > 1 \text{ or } \beta = \infty \implies \sum a_n \text{ diverges} \\ \bullet\ \beta = 1 \implies \text{no conclusion.} \end{cases}$

Example

Do the following series converges?

$$(a)\sum_{n=0}^{\infty}\frac{3^n+7}{5^n}, \qquad (b)\sum_{n=1}^{\infty}\frac{n}{3^n}.$$

Solution.

(a) Let $a_n=\dfrac{3^n+7}{5^n}$. Consider the ratio

$$\frac{a_{n+1}}{a_n}=\frac{3^{n+1}+7}{3^n+7}\cdot\frac{5^n}{5^{n+1}}\to\frac{3}{5}<1,\ \text{as } n\to\infty.$$

By the Ratio Test, (a) converges.

(b) Let $a_n=\dfrac{n}{3^n}$. Consider the root

$$\sqrt[n]{a_n}=\sqrt[n]{\frac{n}{3^n}}=\frac{\sqrt[n]{n}}{3}\to\frac{1}{3}<1,\ \text{as } n\to\infty.$$

By the Root Test, (b) converges.

Definition (Absolute convergence)

Let (a_n) be a sequence of real numbers. A series $\sum_{n=1}^{\infty} a_n$ is said to be absolutely convergent if the series

$$\sum_{n=1}^{\infty} |a_n| = |a_1| + |a_2| + \cdots + |a_n| + \cdots$$

is convergent.

Since $|a_n| \geq 0$, for every $n \geq 1$, we may apply various convergence tests for non-negative series to the series $\sum_{n=1}^{\infty} |a_n|$.

• The Ratio & Root Tests for arbitrary series

Consider limits $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \alpha$, $\lim_{n\to\infty}\sqrt[n]{|a_n|} = \beta$.

1. $\begin{cases} \bullet\ \alpha < 1 \implies \sum a_n \text{ is converges absolutely} \\ \bullet\ \alpha > 1 \text{ or } \alpha = \infty \implies \text{both } \sum |a_n| \text{ and } \sum a_n \text{ diverge} \\ \bullet\ \alpha = 1 \implies \text{no conclusion.} \end{cases}$

2. $\begin{cases} \bullet\ \beta < 1 \implies \sum a_n \text{ is converges absolutely} \\ \bullet\ \beta > 1 \text{ or } \beta = \infty \implies \text{both } \sum |a_n| \text{ and } \sum a_n \text{ diverge} \\ \bullet\ \beta = 1 \implies \text{no conclusion.} \end{cases}$

Example

Is the series $\sum_{n=1}^{\infty}(-1)^n\frac{n}{3^n}$ absolutely convergent?

Solution. Let $a_n = (-1)^n\frac{n}{3^n}$. Then $|a_n| = \frac{n}{3^n}$.

By either the Ratio Test

$$\lim_{n\to\infty}\frac{|a_{n+1}|}{|a_n|} = \lim_{n\to\infty}\frac{n+1}{3^{n+1}}\cdot\frac{3^n}{n} = \frac{1}{3}\lim_{n\to\infty}\frac{n+1}{n} = \frac{1}{3},$$

or the Root Test

$$\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\sqrt[n]{\frac{n}{3^n}} = \lim_{n\to\infty}\frac{\sqrt[n]{n}}{3} = \frac{1}{3}\lim_{n\to\infty}\sqrt[n]{n} = \frac{1}{3},$$

the series $\sum\frac{n}{3^n}$ converges, and hence the given series converges absolutely.

• Absolute convergence $\Longrightarrow$ convergence

If a series $\sum a_n$ is absolutely convergent, then it is convergent.

In other words, if $\sum |a_n|$ converges, then $\sum a_n$ converges.

But if $\sum |a_n|$ does not converge, then $\sum a_n$ may or may not converge.

Definition (Conditional convergence)

A series $\sum a_n$ is said to be conditionally convergent if it is convergent but not absolutely convergent. That is, $\sum a_n$ converges but $\sum |a_n|$ diverges.

For example, the alternating harmonic series $\sum \frac{(-1)^{n-1}}{n}$ is not absolutely convergent, but it is conditionally convergent.

Example

The alternating p-series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p} \quad (p > 0)$$

converges by the alternating series test.

This series

- absolutely converges for $p > 1$ (because $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges),
- conditionally converges for $0 < p \leq 1$ (because $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges).

• Summary

To determine whether a series converges absolutely or conditionally, we do the following steps.

1. **First**: Check for absolute convergence, i.e, to investigate whether the series $\sum |a_n|$ converges.
2. **Next**:
   - If YES, then the given series $\sum a_n$ converges absolutely;
   - If NO, then the given series $\sum a_n$ does not converge absolutely, we proceed to determine whether the given series $\sum a_n$ is convergent or divergent.